

Dr Xiaoyu Tong

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Education

- PhD** **University of Amsterdam**, PhD in Natural Language Processing – Amsterdam, NL July 2026
- Thesis: *Dissecting Incongruity: Metaphor and Humor Understanding of Large Language Models*
 - Supervisors: Dr Ekaterina Shutova, Dr Martha Lewis
- MSc** **University of Amsterdam**, MSc in Brain and Cognitive Sciences – Amsterdam, NL Mar 2021
- Thesis: *Metaphor Paraphrasing and Word-Sense Disambiguation: Toward a New Approach to Automated Metaphor Processing*
 - Supervisors: Dr Martha Lewis, Dr Ekaterina Shutova
- MA** **Xiamen University**, MA in Linguistics – Xiamen, China July 2017
- Thesis: *The Writer's Operation of Extended Metaphor in Meaning Construction*
 - Supervisor: Prof. Zhiying Xin
- BA** **Xiamen University**, BA in English – Xiamen, China July 2014
- Minor in Chinese Linguistics

Research Experience

- Independent Researcher**, Independent Researcher – Remote July 2026 – present
- Research on LLMs' cultural alignment in humour processing, collaboration with Dr Ekaterina Shutova and Dr Martha Lewis at University of Amsterdam
 - Research on application of LLMs in speech-language therapy for people with aphasia, collaboration with Dr Jie Fu at Xingtai Medical College
- Faculty of Science, University of Amsterdam**, Doctoral Researcher – Amsterdam, NL Sept 2021 – July 2026
- Conducted interdisciplinary research on evaluating LLMs' metaphor, humour, pragmatic, and cultural understanding in text and multimodal communication.
- Created theory-driven benchmarks and evaluation protocols for testing linguistic, pragmatic, and cultural capabilities in unimodal and multimodal LLMs.
 - Designed annotation schemes grounded in cognitive and linguistic theories, achieving high reliability (Gwet's AC1 0.84).
 - Led cross-cultural behavioural studies involving 42K human responses across 4K textual and multimodal stimuli.
 - Benchmarked frontier open- and closed-source models using prompting and fine-tuning approaches.
 - Combined statistical analysis, error analysis, and qualitative interpretation to compare human and model behaviour.
- Institute for Logic, Language and Computation, University of Amsterdam**, Research Intern May 2020 – Mar 2021 – Amsterdam, NL
- Reframed a widely used, theory-driven manual protocol for metaphor identification (MIPVU) as a word-sense disambiguation (WSD) problem, and operationalised this insight in a novel dataset and evaluation framework for automated metaphor interpretation.
 - Created a resource that led to the BIG-bench metaphor-understanding benchmark and MUNCH project.

Funding

- China Scholarship Council Doctoral Scholarship (2021 – 2025)

Teaching and Mentoring

- Thesis Mentor, MSc Logic, University of Amsterdam (2023)
- Teaching Assistant, Introduction to the Philosophy of Language, MSc Logic, University of Amsterdam (2022)
- Teaching Assistant, Introduction to Linguistics, BA English, Xiamen University (2015 – 2017)

Service

- Reviewer, ACL Rolling Review (2024 – present)
- Reviewer, *IEEE Transactions on Audio, Speech and Language Processing* (2026)
- Reviewer, *Journal of Cognitive Psychology* (2025)
- Reviewer, NAACL 2024 FigLang Workshop (2024)
- Member, Institute for Logic, Language and Computation (ILLC) PhD Council (2022 – 2023)

Publications

- Evaluating LLMs’ cultural alignment in humor appreciation** present
Xiaoyu Tong, Ivo Verhoeven, Martha Lewis, Ekaterina Shutova
(In Preparation)
- Dissecting Incongruity: Metaphor and Humor Understanding of Large Language Models** 2026
Xiaoyu Tong
hdl.handle.net/11245.1/0bfb1d70-c5e8-4538-a554-d4c0e2ec0980 (University of Amsterdam)
- Hummus: A Dataset of Humorous Multimodal Metaphor Use** 2026
Xiaoyu Tong, Zhi Zhang, Pia Sommerauer, Martha Lewis, Ekaterina Shutova
10.48550/arXiv.2504.02983 (Conditionally accepted by TACL)
- A framework for annotating and modelling intentions behind metaphor use** 2026
Gianluca Michelli, **Xiaoyu Tong** (co-first author), Ekaterina Shutova
10.18653/v1/2026.starsem-conference.4 (*SEM 2026)
- Metaphor Understanding Challenge Dataset for LLMs** 2024
Xiaoyu Tong, Rochelle Choenni, Martha Lewis, Ekaterina Shutova
10.18653/v1/2024.acl-long.193 (ACL 2024)
- Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models** 2023
Aarohi Srivastava, Abhinav Rastogi, ... **Xiaoyu Tong**, ... Ziyi Wu
openreview.net/forum?id=uyTL5Bvosj (TMLR, Featured)
- Recent advances in neural metaphor processing: A linguistic, cognitive and social perspective** 2021
Xiaoyu Tong, Ekaterina Shutova, Martha Lewis
10.18653/v1/2021.naacl-main.372 (NAACL 2021)

Released Benchmarks and Datasets

- Hummus: Dataset and benchmark of humorous multimodal metaphor use** 2026
github.com/xiaoyuisrain/humorous-multimodal-metaphor-use
- MUNCH: Metaphor Understanding Challenge Dataset** 2024
github.com/xiaoyuisrain/metaphor-understanding-challenge
- Co-created dataset of intentions behind metaphor use** 2024
github.com/GMichelli/intentions-behind-metaphor
- BIG-bench metaphor understanding benchmark task** 2021
github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks/metaphor_understanding

Skills

Programming and research computing: Python, R, Bash, JavaScript, HTML/CSS, Git/GitHub

LLM and VLM research: Hugging Face Transformers, PyTorch, PEFT; open-source model experimentation on remote high-performance computing (HPC) infrastructure; programmatic API-based evaluation of proprietary models; prompt engineering, fine-tuning, and reproducible evaluation workflows

Dataset creation and research data management: dataset and annotation scheme design; JSON and other structured data formats; web data extraction

Experimental design, data collection, and annotation: Qualtrics, Prolific, Amazon Mechanical Turk, Labelbox; web-based behavioural experiment and annotation interfaces; API-based study deployment, participant recruitment, and data collection

Data processing and statistical analysis: reproducible data-processing pipelines; data cleaning, transformation, quality control, and validation; regression modelling and statistical inference; publication-quality data visualisation

Research communication and publication: LaTeX, Inkscape

Languages

Mandarin Chinese: Native

English: Full professional proficiency, academic working language

Japanese: Proficient reading; strong conversational listening and speaking; limited formal writing

Interests

Artificial Intelligence: Evaluation, Human-AI Interaction, Human-Centered AI, LLM, World Model

Linguistics: Figurative Language, Humour, Pragmatics, Corpus Linguistics

Cognitive Science: Embodied Cognition, Conceptualisation, Categorisation